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Multi-magnification mask R-CNN and HED-based segmentation with hybrid DTCWT-MSER feature extraction and optimized dual-classifiers for automated colon biopsy malignancy grading

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Abstract

Biopsy images from the colon are necessary to discover cancer, since they display particular layouts of biological elements. Histopathologists use a microscope to inspect the appearance of cells since diagnosing cell deformation is hard and opinion-based. Since opinions differ among human viewers, a sample may receive different cancer grades from one pathologist to another or even from the same person. Consequently, more people are interested in making automated systems for grading images from colon biopsies. Dependable identification at 4X, 10X, and 40X by this approach fostered various exciting developments in computerized examination of colon cells. The authors also highlight a segmentation process that works at any magnification by joining Mask R-CNN with the HED network. Moreover, the system relies on contrast enhancement and color normalization to ensure all cells look the same, making it easier to use features to classify them. Next, hybrid features are created using texture analysis from DT-CWT, deep features from fully connected convolutional networks, and information from morphology and MSER. A set of HOG characteristics is included in the strong features used, providing a more detailed description for the exact identification of cells. Improved Snake Optimization is applied to select the significant features, which boosts the computer's performance. To summarize, the system takes advantage of the dual-classifier idea by using both RF and LSTM to identify uncommon and repeated data patterns. The proposed fusion of features was found to perform consistently well on the different datasets. Using the AMC and Imediatreat datasets, the model obtained a very high accuracy score and its F1-score reached 98.70%. Despite not achieving high precision, the model performed well on the IPC dataset. Furthermore, the findings from GlaS prove that the model works well, with a precision of 95.00% and an accuracy of 95.50%. Overall, the outcomes point to the feasibility of the model on other data with its high rate of detection and diagnoses.

Keywords CNN · DT-CWT · HED network · HOG · LSTM · Mask R-CNN · MSER · Random forest

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1 Introduction

Microscopic imaging is vital for examining biopsy slides, and medical imaging has become indispensable for identifying and forecasting several illnesses. These images illustrate how the tissues and cellular constituents of various body sections are arranged in distinct patterns. For instance, distinct spatial configurations of biological elements can be observed in colon biopsy images used here to diagnose cancer. Histopathologists often diagnose colon cancer by visually inspecting the tissue deformation under a microscope and determining if the sample tissues' geometric structure and organization seem normal or malignant. But because cancer grades are frequently determined by the observed tissue shape, which results in diversity between and within observers, this procedure is both subjective and time-consuming. For the same sample, different histopathologists—or even the same person—may provide varying cancer grades.

Researchers have been creating automated, quantitative methods to assess tissue deformation and objectively grade colon biopsy samples for malignancy in order to overcome these constraints. Research on the detection of colon cancer has advanced in several ways. To categorize samples into several classes, for example, some researchers have used hyperspectral data from colon biopsies, choosing particular spectral bands and computing characteristics. Others have used methods like laser-induced fluorescence and Raman spectroscopy to look at genetic changes or variations in blood serum composition as biomarkers for colon cancer.

Analysis of textural differences in normal and malignant colon biopsy images has also been a major area of research. By adding fractal dimensions, the authors of [1] showed how texture characteristics Like entropy and correlation may be used to categorize biopsy images with an accuracy of up to 94.27%. By computing morphological characteristics and gray-level co-occurrence matrix (GLCM) features, the authors of [2] expanded on this work and used CNN classifiers to achieve classification accuracies of up to 90%. More recently, the authors of [3] presented a hybrid classification approach for colon biopsy images that used statistical moments, Haralick texture characteristics, and Histogram of Oriented Gradients (HOG) features. This approach produced an amazing 98.85% accuracy rate using an ensemble SVM classifier.

Some restrictions still exist in spite of these developments. For instance, many texture-based methods do not completely utilize certain morphological information on colon tissue structure, while graph-based methods are computationally costly. These difficulties highlight the necessity of a computer-aided diagnosis system that combines texture-based and morphological aspects to improve diagnostic precision while maintaining computational efficiency. By addressing the subjective aspect of conventional histological investigation, such a method might greatly increase the accuracy and efficiency of colon cancer diagnosis. The differences in complexity, size, and tissue structure across various magnification levels make it difficult for computer vision to analyze colon tissue at different magnifications, especially at 4X, 10X, and 40X. Different aspects are shown by each magnification: 40X enables a close-up look at individual cells, 10X concentrates on certain cell groups, and 4X offers a more comprehensive tissue background. It is necessary to use segmentation and feature extraction techniques that are specific to each magnification to handle these variances. Specialized machine learning (ML) and deep learning (DL) techniques are necessary to get precise categorization and detection in colon tissue examination across various magnifications. Multi-scale segmentation is a crucial method that uses algorithms that dynamically adjust to various magnification levels to separate areas of interest (ROIs) precisely according to scale-specific features. Higher magnifications benefit from finer segmentation that accentuates complex cellular features, but lower magnifications could necessitate coarser segmentation to capture the overall tissue structure. Magnification-specific feature extraction is another essential method; at higher magnifications, the emphasis is on fine-grained textural patterns and cell morphology, whereas at lower magnifications, global texture and spatial arrangement characteristics are given priority. Classification accuracy may be further increased by integrating ML and DL models, especially convolutional neural networks (CNNs) [4] that have been trained or optimized for characteristics unique to magnification. Particularly interesting are hybrid methods that allow reliable classification at different magnifications by combining deep, hierarchical characteristics taken from CNNs with manufactured features. According to the above discussion, the research question is: "How does a hybrid segmentation and classification method that relies on both deep learning and artificially created features help ensure accuracy and consistency in automated detection of colon cancer across different histopathology magnification levels?" Across several magnification levels (4X, 10X, and 40X), the suggested technology significantly advances the fields of automated colon tissue analysis and cancer detection. Important contributions consist of:

- Magnification-Independent Segmentation: The method tackles the difficulty of delivering consistent analysis across many magnification levels by adopting a unique hybrid segmentation strategy that merges Mask R-CNN with the Holistically-Nested Edge Detection Network (HED). While Mask R-CNN enables instance segmentation by recognizing individual tissue components and creating pixel-level masks, HED plays a significant role in strengthening border delineation. Specifically, HED refines edge detection for finer delineation of disease-affected areas, leveraging multi-scale edge detection and context-aware refinement to attain precision in capturing complicated and irregular patterns, which are frequent in colon tissue analysis. This dual method allows the algorithm to discriminate between overlapping structures and irregularly shaped cellular components with great accuracy. This flexibility is further supported by Mask R-CNN's Region Proposal Network (RPN) and ROI alignment, which guarantee that comprehensive segmentation may be accomplished reliably at different magnifications (e.g., 4X, 10X, and 40X). Accurately identifying and defining disease characteristics at every size depends on this magnification-independent competence.
- Advanced Preprocessing for Clarity: To improve image clarity at different magnification levels and standardize tissue appearance, preprocessing techniques include color normalization and contrast augmentation. By lowering color and contrast variability, this preprocessing increases the efficiency of segmentation and feature extraction.
- Region of Interest (ROI) and Feature Extraction: A sophisticated hybrid feature extraction procedure is used after segmentation. To capture form, structure, and connectivity within the tissue, morphological characteristics and Maximally Stable Extremal Regions (MSER) extraction techniques are used on the discovered ROIs. Furthermore, a fully connected convolutional network and the Dual Tree Complex Wavelet Transform (DTCWT) are used to extract texture and deep features. Histogram of Oriented Gradients (HOG) features are added to this strong feature set to create a hybrid feature representation that incorporates deep learning, morphological, and spatial information—all essential for precise tissue categorization.

- Feature Selection Optimization: A feature selection approach based on Enhanced Snake Optimization is created in order to improve the extracted features and guarantee computational efficiency. By identifying the most pertinent characteristics, this optimization lowers model complexity and increases classification accuracy.
- Novel Dual-Classifiers: Long Short-Term Memory (LSTM) and Random Forest classifiers are combined in a novel dual-classifier technique. By combining the strong ensemble learning of Random Forests with the temporal sequence learning capability of LSTM, this hybrid classifier improves classification accuracy by identifying both complicated, nonlinear relationships and sequential patterns in the data.

In Section II, the study begins with a survey of a broad variety of literature, emphasizing relevant research works on the topic. The materials and methods are described in Section III. The proposed methodology for this research paper is provided in Section IV. The outcomes of MATLAB-based simulations are then shown in Section V, along with a thorough analysis. Section VI concludes the study by summarizing the results and offering the future scope for the research paper.

2 Literature review

The use of deep neural networks for gland segmentation within histopathology images, namely for colon gland analysis, has advanced significantly in recent years. To categorize individual pixels as belonging to either normal or malignant glands, the authors of [4], for example, developed a deep convolutional neural network (CNN)-based pixel classifier for the semantic segmentation of colon gland images. This was accomplished by employing two 7-layer CNNs. A weighted total variation normalization technique inside a figure-ground segmentation approach was used to refine these predictions. In a similar vein, the authors of [5] created a CNN model specifically tailored for gland segmentation that integrates radiomic information with deep learning and conventional radiomic characteristics via an SVM classifier, improving segmentation accuracy. By using a multi-task learning framework that integrates multi-level contextual information and auxiliary supervision to avoid vanishing gradients, the authors of [6] expanded on this idea with a contour-aware network for gland segmentation.

Additionally, by reintroducing the original image at various places along the network, the authors of [7] suggested a CNN to compensate for the information loss caused by max-pooling. This network used random changes to enhance segmentation resilience, including the creation of an uncertainty map to pinpoint confusing regions, used varied dilation rates for multi-level feature aggregation, and utilized atrous spatial pyramid pooling to preserve image resolution.

Building on these developments, our suggested gland segmentation method seeks to detect interior glandular structures by utilizing their geometrical and morphological characteristics, as well as to define gland borders. Different radiomic approaches have been successful in differentiating between benign and malignant colon lesions in terms of categorization. The authors of [8] employed local binary patterns in conjunction with a Gaussian SVM to attain a respectable classification accuracy, which was further improved using ensemble classifiers and hybrid feature sets. Improvements in classification results have also been demonstrated when textural analysis of color components is combined with Histogram of Oriented Gradients (HOG) features. By suggesting radiomic pipelines to evaluate colorectal cancer using shape and texture characteristics with different classifier models, the authors of [9] have made a further contribution. CNN-based models outperform conventional classifiers in terms of accuracy.

Apart from segmentation, graph-based techniques have also been investigated for colon cancer grading [10]. For instance, in order to obtain high accuracy in cancer identification and grading, the authors of [11] used a circle recognition algorithm to cluster pink, purple, and white areas in biopsy images. They then created graphs based on these circular clusters. Similarly, the authors of [12] used graph-based methods to evaluate malignancy by contrasting biopsy image test graphs with reference graphs of healthy tissues. While Awan et al. suggested the "Best Alignment Metric" (BAM) to increase gland shape analysis and provide reliable grade classifications.

Despite advancements in colon cancer detection, grading, and segmentation, there are still not many comprehensive computational pathology pipelines available. By offering a completely automated pipeline that combines gland segmentation, cancer identification, and grading into a single system for thorough histopathological study of colon cancer, our research seeks to overcome this issue. This pipeline will help with the uniform grading of colon cancer, increase diagnostic efficiency, and streamline computational pathology procedures.

The following is a tabular literature review that summarizes the latest developments in deep learning-based techniques for the diagnosis of colon cancer, detailing the methodology, benefits, drawbacks, and conclusions of each study (Table 1):

This table summarizes diverse approaches, indicating that while deep neural networks and hybrid models improve diagnostic accuracy, challenges like computational cost, dependency on specific datasets, and feature extraction complexity persist.

Research gaps Despite the promising advancements in automated cancer detection using deep learning and machine learning techniques, several critical gaps persist in the existing literature. One significant challenge is magnification variability, where many methods struggle to maintain accuracy across different magnification levels. This inconsistency can lead to varying results in cancer detection, which is crucial for reliable diagnosis. Additionally, the issue of complex feature extraction arises, as current approaches often rely heavily on either deep learning features or handcrafted features, lacking effective integration of both methodologies. This limitation can hinder the extraction of meaningful representations from histological images.

Furthermore, there is a pressing need for more efficient feature selection and optimization methods that can reduce dimensionality while preserving relevant information. This enhancement is vital for improving the computational efficiency of the models without sacrificing accuracy. Additionally, the problem of class imbalance remains unaddressed in. several existing methods. When datasets are imbalanced, models may exhibit biased predictions toward the more frequent classes, compromising the reliability of the cancer detection system.

Table 1 Tabular comparison for literature review

Reference	Methodology	Advantages	Limitations	Outcome	Accuracy
[13]	Pre-trained SegNet, UNet, and ResNet models applied to the AiCOLO dataset for colon cancer segmentation.	High accuracy with deep model integration.	Requires large datasets for training.	ResNet achieved 96.98% accuracy.	96.98%
[14]	Deep learning segmentation model for abnormal cell detection, utilizing annotations for diagnostic assistance.	Generates informative annotations aiding physicians.	Limited scope of features for complex tissue analysis.	Efficient abnormal cell detection for colon cancer.	93.48%
[15]	Pre-trained ResNet model for colon cancer diagnosis, comparing ResNet50 with ResNet18.	High sensitivity and better performance with ResNet50.	Limited by dataset division, may affect generalizability.	ResNet50 achieved 87% sensitivity and 80% accuracy.	86.50%
[16]	WSI image classification using different magnification factor labels for colorectal cancer grading.	Effective grading across magnification levels.	Limited to specific WSI image datasets.	Achieved 94.6% accuracy in cancer grading.	94.60%
[17]	Deep learning network for segmentation of H&E-stained colon cancer images.	Early diagnosis potential with accurate segmentation.	Requires specific stain types; limited generalizability.	Achieved 94.8% accuracy for tumor tissue segmentation.	94.80%

Proposed solutions The proposed system aims to address these gaps through several innovative strategies. First, it introduces magnification-independent segmentation by integrating Mask R-CNN with the Holistically-Nested Edge Detection (HED) network. This integration allows the system to achieve consistent segmentation across multiple magnification levels (4X, 10X, and 40X), enhancing boundary delineation and accurately capturing complex tissue patterns. Such a dual approach is essential for reliably identifying disease-affected regions.

To further improve analysis, the system incorporates advanced preprocessing techniques, such as color normalization and contrast enhancement. These steps standardize tissue appearance, thereby enhancing the effectiveness of segmentation and feature extraction processes. The proposed system also employs a hybrid feature extraction strategy that combines various techniques, including morphological features, Maximally Stable Extremal Regions (MSER), Dual Tree Complex Wavelet Transform (DTCWT), and Histogram of Oriented Gradients (HOG). This comprehensive feature set captures both spatial and deep learning features, significantly enhancing classification accuracy. Additionally, the system implements a feature selection optimization approach using the ESO algorithm. This algorithm selectively identifies the most relevant features, thereby improving computational efficiency and classification performance. The proposed solution also features a dual-classifier approach, combining Long Short-Term Memory (LSTM) and Random Forest classifiers. This hybrid classification strategy effectively captures both sequential patterns and complex relationships within the data, thereby improving overall classification accuracy. Our approach differs from other studies because it can be used at any magnification, since it combines Mask R-CNN with the Holistically-Nested Edge Detection (HED) network. Using two optical networks means that every scale of magnification will get boundaries that are set consistently, which usually corrects small errors that happen in multiscale processing. Besides, previous models depend on deep learning or homegrown features, but our method is different, as it brings together DTCWT, HOG, MSER, morphological descriptors, and deep features. Because of the hybrid feature extraction, the representation is more descriptive and helps achieve much better classification performance. To solve problems with large amounts of data, we make use of ESO to help determine important features. It is better than previous methods, such as PCA, as it reduces repetition, maintains useful features, boosts the accuracy of the model, and helps improve how computers respond to this kind of data. Additionally, we find that it is important to standardize input, so we preprocess it with techniques such as balancing colors and increasing contrast. As a result, the same levels of consistency and reliability can be obtained when staining and imaging vary in real life. All in all, the chosen system achieves significant progress in the sector due to its adaptable approach, multiscale use of various models, effective optimization, and advanced preprocessing methods. All this improves the dependability, sizes, and accuracy of automated systems for detecting colon cancer in health care, handling problems found in recent studies.

3 Materials and methods

3.1 Enhanced snake optimization (ESO)

The ESO algorithm is a metaheuristic technique inspired by the mating behavior of snakes, proposed by the authors of [18]. ESO is designed for solving optimization problems by iteratively updating a population of solutions represented as real-number vectors. Each iteration of the algorithm involves the following steps:

1. Initialization: The population of solutions is initialized randomly.
2. Perturbation: A new solution is generated by perturbing one of the existing solutions as follows:

$$x_t = x_i + \epsilon \tag{1}$$

Where:
x_t is the new solution.
x_i is the existing solution.
ϵ is a random vector.

3. Fitness Evaluation: The fitness of the new solution is determined using the objective function:

$$f\left(x_t\right) \tag{2}$$

a. Opposition-Based Learning: ESO employs a novel opposition-based learning strategy to update existing solutions. This strategy involves creating a new solution that is the opposite of the existing solution in the search space. The new solution is then evaluated, and if it has superior fitness compared to the existing solution, the existing solution is replaced by the new one.
b. Stopping Criterion: The algorithm terminates when a predefined stopping criterion is met. Common criteria include reaching a maximum number of iterations or achieving a minimum fitness value.

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The ESO algorithm demonstrates effectiveness in tackling a wide range of optimization problems, including function optimization, scheduling, and image processing.

Algorithm

Initialization: Start by initializing a population of solutions randomly.

For Each Iteration:

1. **Create a New Solution by Perturbing an Existing Solution:**
   - Select an existing solution, x_i.
   - As per equation (1), generate a new solution, x_t, by adding a random vector, ε, to x_i:
2. **Evaluate the Fitness of the New Solution:** Assess the fitness of the new solution, x_t, using the ESO objective function:

$$f(x_t) = f_1(x_t) + f_2(x_t) \tag{3}$$

   Where:
   - $f_1(x_t)$ represents a fitness measure of solution quality.
   - $f_2(x_t)$ represents a fitness measure of solution diversity.
3. **Update Existing Solutions Using Opposition-Based Learning:** For each existing solution, x_i:
   - Create a new solution, x_o, by taking the opposite direction of x_i in the search space:

$$x_o = -x_i \tag{4}$$

   - If the fitness of the new solution, $f(x_o)$, is better than the fitness of x_i, replace x_i with x_o.
4. **Repeat Steps 2-3 Until a Stopping Criterion is Met:** Continue iterating through steps 2 and 3 until a predefined stopping criterion is achieved. This criterion can include reaching a maximum number of iterations or achieving a specified fitness threshold.

4 Proposed methodology

The proposed colon cancer detection framework's flow diagram is displayed in Fig. 1, which includes many crucial steps for processing and categorizing histopathological images.

High-resolution images are first obtained from the database, and then image pre-processing techniques, including contrast enhancement and color normalization, are applied. Then, segmentation is carried out using a hybrid model that fuses the outputs for refined regions of interest and combines an improved HED network for better edge detection with Mask R-CNN for instance segmentation. Then, utilizing morphological statistical, MSER, DT-CWT + CNN, and HOG features, feature extraction is carried out to capture distinctive attributes. These features are then integrated into a single feature vector. In order to produce accurate predictions, the feature set is refined through the use of ESO for feature selection. A dual-classifier architecture consisting of Random Forest and LSTM classifiers is then used for classification. The next sections include the complete approach.

4.1 Image acquisition from the database

In this study, the first essential step is obtaining images from a pre-established histopathological image database, specifically designed for colon tissue analysis. The database comprises images captured at multiple magnification levels, denoted by scales $\mathcal{M}$ (e.g., 4X, 10X, and 40X). These images are maintained in a standardized color format to ensure consistency across analyses and are often stored as high-resolution RGB arrays with supplementary metadata.

Each image is represented by a three-dimensional matrix $\mathbb{I}$ indexed as $\mathbb{I}_{ijk}$, where i and j correspond to spatial coordinates (height and width of the image), and k denotes the color channel, such that:

$$\mathbb{I} = \left\{ I_{ijk} \in R^{+} | i \in [1, H], j \in [1, W], k \in \{R, G, B\} \right\} \tag{3}$$

Here, H and W are the image dimensions in pixels, and $\{R, G, B\}$ specifies the red, green, and blue color channels, respectively, with I_{ijk} representing the intensity value at a given pixel.

The selection of a specific image $\mathbb{I}_n$ from the database is based on its associated metadata attributes, such as patient ID $\mathcal{P}$, diagnostic category $\mathcal{D}$, and magnification $\mathcal{M}$. An image retrieval function, $\mathcal{Q}(\mathcal{P}, \mathcal{D}, \mathcal{M})$, is defined to extract an image from the dataset as:

$$\mathcal{Q}(\mathcal{P}, \mathcal{D}, \mathcal{M}) \rightarrow \mathbb{I}_n \tag{4}$$

Once retrieved, the image $\mathbb{I}_n$ is preloaded for further steps in the pipeline, where it will be processed for segmentation, feature extraction, and analysis. This standardized retrieval and acquisition approach ensures the integrity and uniformity of data handling throughout the research work.

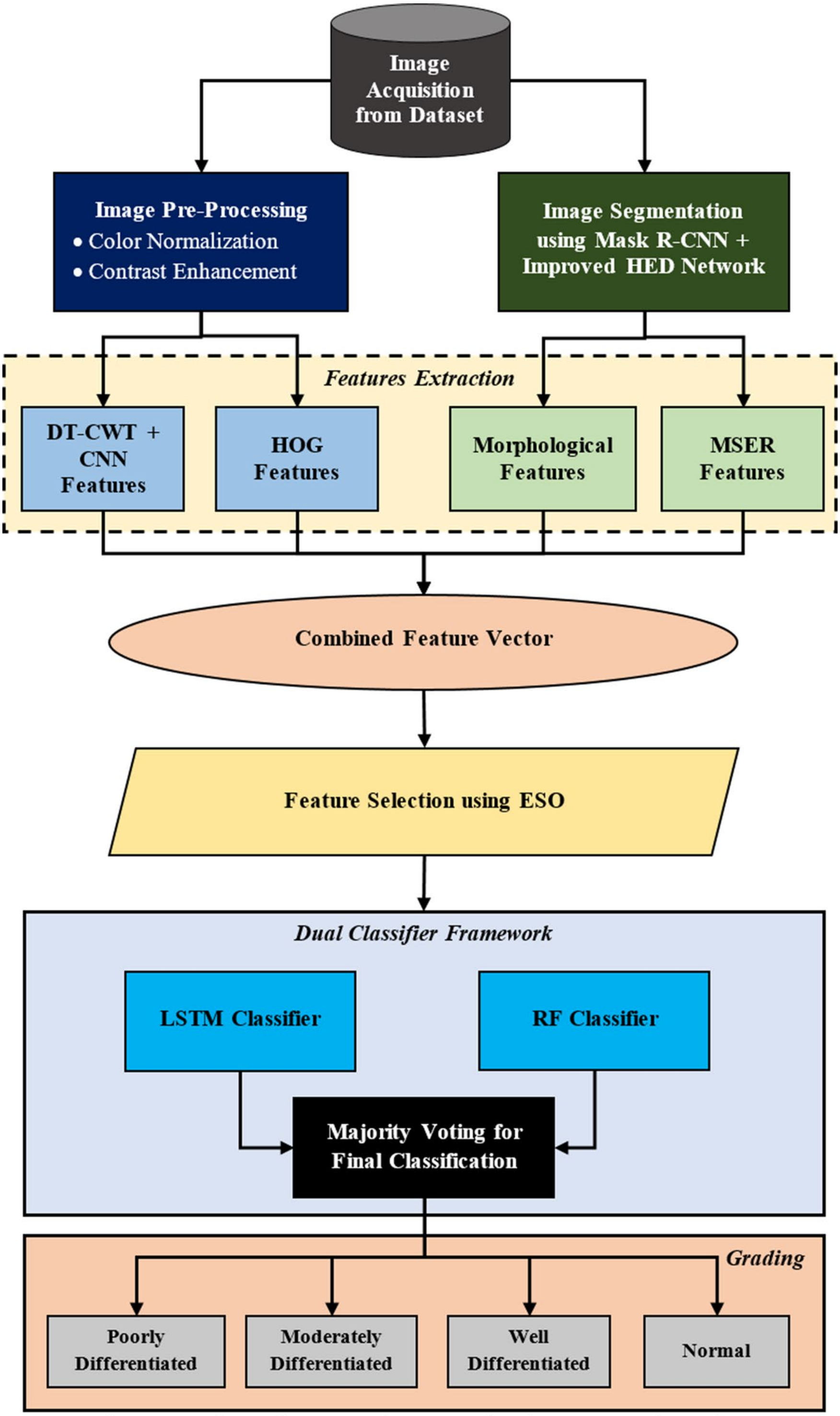


Fig. 1 Flow diagram of proposed colon cancer detection

4.2 Image pre-processing

The pre-processing stage is essential for preparing colon biopsy images, ensuring uniformity across varied imaging conditions and enhancing the visual clarity required for subsequent analysis. This phase encompasses two primary techniques: Color Normalization and Contrast Enhancement [19]. Each step is meticulously designed to address variations in image appearance, facilitating a consistent foundation for feature extraction and segmentation.

4.2.1 Color normalization

Due to inconsistencies in staining intensities and lighting conditions during image acquisition, colon tissue images often exhibit color variations that can hinder analysis accuracy. To counter this, a normalization process is applied, aligning the color distributions of all images to a common reference standard. This process adjusts each color channel individually, creating a balanced appearance across all images in the dataset.

In this formulation, let each image pixel at position (x, y) be represented in the RGB color space as $P(x, y) = (R(x, y), G(x, y), B(x, y))$. Normalized intensity values for each channel are computed with reference to a target image P_{ref} as follows [19]:

$$\begin{array}{c} R_{norm}(x, y) = \alpha_R \times R(x, y) + \beta_R \\ G_{norm}(x, y) = \alpha_G \times G(x, y) + \beta_G \\ B_{norm}(x, y) = \alpha_B \times B(x, y) + \beta_B \end{array} \tag{5}$$

Where α and β are scaling and offset coefficients respectively, determined by matching the histogram statistics of each channel to the reference. The resulting normalized image P_{norm} thus aligns with the reference image's color characteristics, reducing variability due to staining differences.

4.2.2 Contrast enhancement

Following normalization, a contrast enhancement procedure is applied to improve the visibility of cellular structures within the images. This step adjusts intensity levels, particularly in grayscale, to create clearer distinctions between foreground tissues and the background, aiding in the identification of critical features.

The Contrast-Limited Adaptive Histogram Equalization (CLAHE) method is employed here, which divides the grayscale image into a grid of $m \times n$ regions (or tiles) and applies localized histogram equalization to each tile. For each tile centered at (i, j), the contrast-enhanced intensity $I_{enh}(i, j)$ is computed as [19]:

$$I_{enh}(i, j) = \min\left(\gamma \cdot \frac{I(i, j) - I_{min}}{I_{max} - I_{min}}, \lambda\right) \tag{6}$$

Where I_{min} and I_{max} represent the minimum and maximum intensities within the tile, γ is a contrast factor, and λ is a clip limit to prevent over-amplification in uniform regions. This adaptive method thus preserves local contrast while ensuring that overly bright or dark regions remain controlled.

After processing, each tile is recombined to form the final contrast-enhanced image. This pre-processed image, with improved consistency in color and contrast, provides a stable and clear representation, laying the groundwork for accurate segmentation and analysis in the subsequent stages.

4.3 Segmentation using hybrid mask R-CNN combined with improved HED network

Mask R-CNN and HED were chosen because they complement each other in resolving the main issues of working with histopathological images. Mask R-CNN does an excellent job at separating intricate, mixed-overlap tissues and can be used at various magnification stages for searching for ROIs. Even so, it could lead to approximate outlines in places with soft characteristics. It addresses this issue by using HED, since it can find thin, fine lines at close range due to deep, multi-scale guidance, which helps highlight the borders between glands and epithelium. The way this approach improves on previous methods is by narrowing the boundaries and improving on object detection with Mask R-CNN.

Following image acquisition, segmentation is performed on each retrieved image $\mathbb{I}_n$ using a hybrid method that combines Mask R-CNN and an enhanced HED network. This hybrid segmentation model is tailored to colon tissue analysis, leveraging Mask R-CNN's instance segmentation capabilities in tandem with HED's refined boundary delineation.

4.3.1 Mask R-CNN for instance segmentation

Mask R-CNN provides pixel-level instance segmentation by identifying individual regions within $\mathbb{I}_n$, generating masks that label specific tissue components. This process starts with the Region Proposal Network (RPN), which generates candidate regions of interest (ROIs), denoted by R_k, where k is the index for each proposed region [20].

For a given image $\mathbb{I}_n$, the RPN produces a set of bounding boxes $B_k = (x_{k1}, y_{k1}, x_{k2}, y_{k2})$, representing the coordinates of each ROI R_k. These ROIs are refined through the ROI alignment process to ensure spatial consistency across different magnification levels $\mathcal{M}$:

$$F_{R_k} = \text{Align}\left(F_{base}, B_k\right) \tag{7}$$

Where F_{base} is the base feature map extracted from $\mathbb{I}_n$, and Align represents the alignment function applied to each bounding box, ensuring accurate feature extraction across spatial coordinates.

From each aligned ROI R_k, a binary mask $\mathcal{M}_k$ is generated by classifying each pixel p within the region:

$$\mathcal{M}_k\left(p\right) = \underset{c}{\text{argmax}}\mathcal{P}\left(c|F_{R_k}\left(p\right)\right) \tag{8}$$

Here, c denotes the possible class labels (e.g., tissue structures), and $\mathcal{P}\left(c|F_{R_k}\left(p\right)\right)$ is the class probability at pixel p given the aligned feature map F_{R_k}.

Figure 2 shows the architecture of Mask R-CNN.

Several magnification-related alterations are included to keep the quality constant at any level (such as 4X, 10X, 40X).

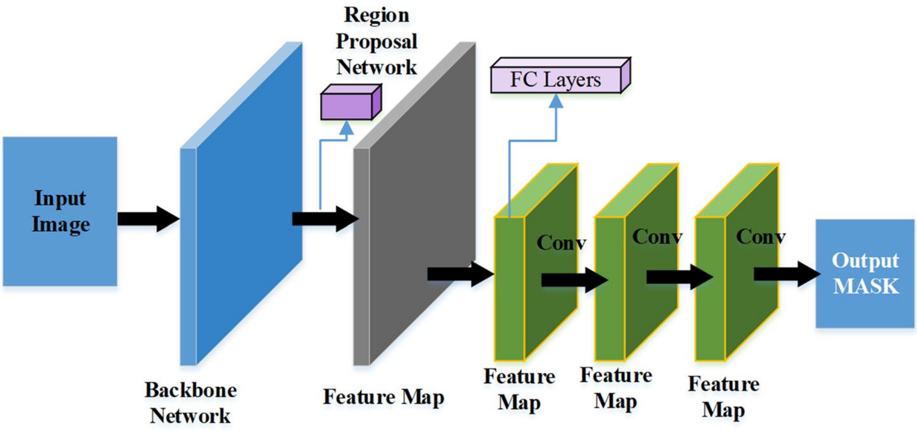


Fig. 2 Architecture of Mask R-CNN

1. Anchor Box Scaling: According to the magnification, the RPN adjusts the size of the anchors. Macular anchors are better suited for low magnification, and smaller anchors are used for high magnification, where tiny structures such as the nuclei can be seen.
2. Training with Multi-Magnification Data: The model is taught using images taken at several magnifications, which makes it resistant to changes in size. The model can still generalize when you change the video recording's zoom level.
3. Feature Normalization: Before analyzing, image processing techniques are used to ensure there is similar color and contrast among the images taken at any scale.
4. Hybrid Segmentation with HED: Along with Mask R-CNN, we use the HED network to help determine better outlines. At high magnification, HED enhances the clarity of the edges around a nucleus, making it easier to outline small and oddly shaped ones.

For persistence in the results at 4X, 10X, and 40X magnification, we make suitable adjustments using the magnification factor in each case.

- With 4X magnification, the tissue looks larger, and its structures appear scattered. Due to this, the RPN in RetinaNet uses large anchor boxes such as 128×128, 256×256, and 512×512 to take in more of the animal's overall features.
- At a magnification of 10X, glands and epithelial areas are strongly visible. The best alternatives are to use 64×64, 128×128, and 256×256 spaces when deciding on the anchor.
- With a 40X magnification, the nucleus and outlines of the cell cytoplasm can both be seen. To get accurate detection of such details, the developer uses smaller anchors such as 16×16, 32×32, and 64×64.

They were picked so that Mask R-CNN's RPN can fit the size of features at each scale, allowing for better instance segmentation.

4.3.2 Enhanced edge detection with HED network

To improve segmentation boundaries, the HED network [21] is employed to enhance edge precision in $\mathbb{I}_n$. The HED network performs multi-scale edge detection to capture complex contours typical of colon biopsy images. Given the input image $\mathbb{I}_n$, HED generates edge maps E_m at different scales m to capture boundaries with varying levels of detail. At each scale m, the edge map E_m is computed as:

$$E_m(i,j) = h_m(\nabla \mathbb{I}_n(i,j)) \tag{9}$$

Where $\nabla \mathbb{I}_n(i,j)$ represents the gradient of $\mathbb{I}_n$ at spatial coordinates (i,j), and h_m denotes the scale-dependent filtering function applied to enhance edges.

The final edge map E_{final} is obtained by combining multi-scale edge maps:

$$E_{final}(i,j) = \sum{}_m \beta_m E_m(i,j) \tag{10}$$

Where β_m is a weight assigned to each scale m, emphasizing relevant boundary details that contribute to accurate tissue structure delineation.

Figure 3 shows the architecture of improved HED network.

Confidence Thresholding of Edge Maps for Colon Histopathological Images at Different Magnifications.

At different levels of magnification, it becomes clear that the edge of the colon tissue is poorly located. Most details appear fuzzy with 4X, as 40X magnification makes them look

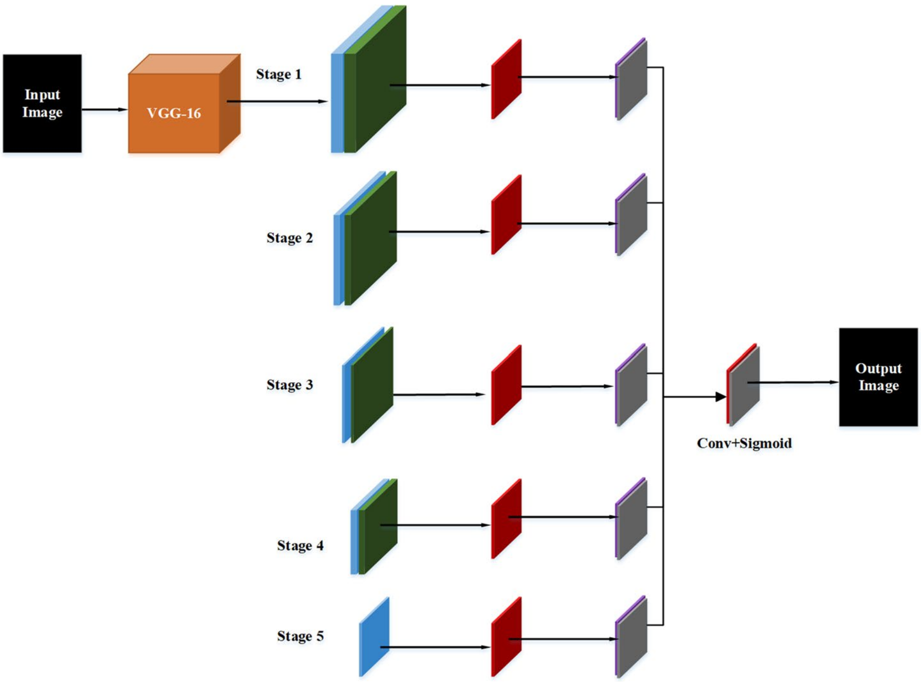


Fig. 3 Architecture of Improved HED-Network

more precise. HED will react differently to different edges, depending on how large the scene is.

Edge probability $E_m(x,y)$, which should be between 0 and 1, is created by the HED model, as long as $m \in \{4x, 10x, 40x\}$ and (x,y) represent the coordinates of a pixel. To make a binary edge map $\widehat{E_m}(x,y)$, the level of confidence τ_m should be set for the chosen magnification.

$$\widehat{E}_m(x,y) = \left\{ \begin{array}{c} 1,\ if\ E_m(x,y) \geq \tau_m \\ 0,\ otherwise \end{array} \right. \tag{11}$$

Threshold values by magnification The value for τ_m varies according to the object's shape and the level of background noise observed at every magnification.

- τ_{40x}=0.4 (higher detail, more confident edges).
- τ_{10x}=0.5 (moderate detail, balanced edge confidence).
- τ_{4x}= 0.6τ4X=0.6 (lower detail, suppress weaker and noisy edges).

Integration into the HED Segmentation Pipeline.

- ***Edge Detection***: The HED model creates a map called $E_m(x,y)$ for every magnification used.
- ***Magnification-Aware Thresholding***: Thanks to τ_m, only the most reliable directions related to the magnification are kept in $\widehat{E_m}(x,y)$.

4.3.3 Fusion of mask R-CNN and HED outputs

The segmentation mask S_{hybrid} is formed by combining instance masks $\mathcal{M}_k$ from Mask R-CNN with the refined edge map E_{final}. For each segmented region R_k, the final hybrid mask S_{hybrid} is defined as:

$$S_{hybrid}(i,j) = \mathcal{M}_k(i,j) \cdot E_{final}(i,j) \tag{12}$$

This formulation ensures that each segmented region within $\mathbb{I}_n$ has precise and clearly defined boundaries, a necessity for identifying complex, overlapping, and irregular cellular structures within colon tissue samples. Figure 4 shows the fusion of Mask R-CNN and the HED Network approach.

4.4 Features extraction

We have gone into detail about the optional steps by discussing the specific role of each method:

- Morphological Features: They are extracted to calculate the shape, size, and connections between different parts of the tissue (e.g., nuclei, glands). Thanks to these features, it is possible to notice unusual growths in tissue that are common in cancer.

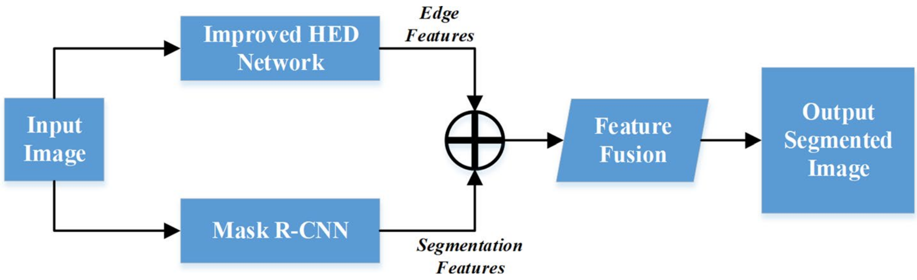


Fig. 4 Segmentation using Hybrid Model

- Maximally Stable Extremal Regions (MSER): With MSER, the high-contrast sections within the ROIs are identified. It helps find out the main parts present in many types of tissues, leading to their identification.
- Dual Tree Complex Wavelet Transform (DTCWT): DTCWT is applied to produce a large number of intelligent and sensitive texture features that reveal the fine changes and broad outlines of an image in different directions and scales. This is necessary for telling apart tissues with minor differences in their structure.
- Fully Connected Convolutional Neural Network (CNN): The CNN is designed to obtain features that show the details in the image, such as the arrangement of colors, various visual elements, and important contexts that are usually difficult to model by hand.
- Histogram of Oriented Gradients (HOG): To highlight the direction of edges and gradients, HOG is used in the network, making it more capable of detecting outlines and contours of both cellular structures and glands for better classification because of their grouping.

If only one type of feature is used in cancer detection from histopathological images, the representation of tissue features may not be complete and might be biased. Thus, combining approaches is the best way to depict the complete features of a tissue's anatomy in images.

To start, characteristics of cellular and tissue structures and their connections are explained using shape and groups of Maximally Stable Extremal Regions (MSER) features. With these features, experts are better able to see any gland changes, uneven cell shapes, and disorder within tissue, all signs that the cells are malignant. On the other hand, they are not able to correctly reflect the differences in fine texture.

Therefore, the DTCWT is used to obtain texture features, while the CNN obtains deep features. They are designed to notice both big and small details, local and international patterns, subtle differences in images, and clues that are isolated areas across an image. In considering DTCWT, it is worth noting that its properties help to distinguish malignant from benign tissue by highlighting important landmarks and matching the same structural features at any size.

Additionally, HOG improves the detection of cellular borders, areas where the epithelial lining is not smooth, and the shape of the glands.

The use of a hybrid representation allows the system to utilize the benefits of both specially designed and deep learning features. As a result, the features become more detailed and separate, which boosts the accuracy of the model when faced with tricky cases and widely varied individuals. Simply put, using hybrid features is necessary to make sure cancer detection can be done accurately across different cases and with more reliability.

Following segmentation and preprocessing, feature extraction is conducted to capture various aspects of tissue morphology and texture, which are essential for analyzing the colon tissue structure in detail. Features are extracted from both the segmented output and the preprocessed image, as shown in Fig. 5 and described in the following sub-sections.

4.4.1 Morphological statistical features

Morphological statistical features provide a quantitative description of the shape and size of segmented tissue regions, which are crucial for identifying structural variations in colon tissue. From the segmented output, S_{hybrid}, generated using the hybrid Mask R-CNN and improved HED network, morphological descriptors are extracted for each identified region R_i.

Each region $R_i \subset S_{hybrid}$ is analyzed for its fundamental morphological attributes [8]:

- **Area** $(\boldsymbol{A_i})$: The area of R_i represents the count of pixels within the region and is calculated as:

$$A_i = \sum\nolimits_{(x,y) \in R_i} 1 \tag{13}$$

Where (x, y) denotes the pixel coordinates. The area metric provides an essential size measurement of cellular components.

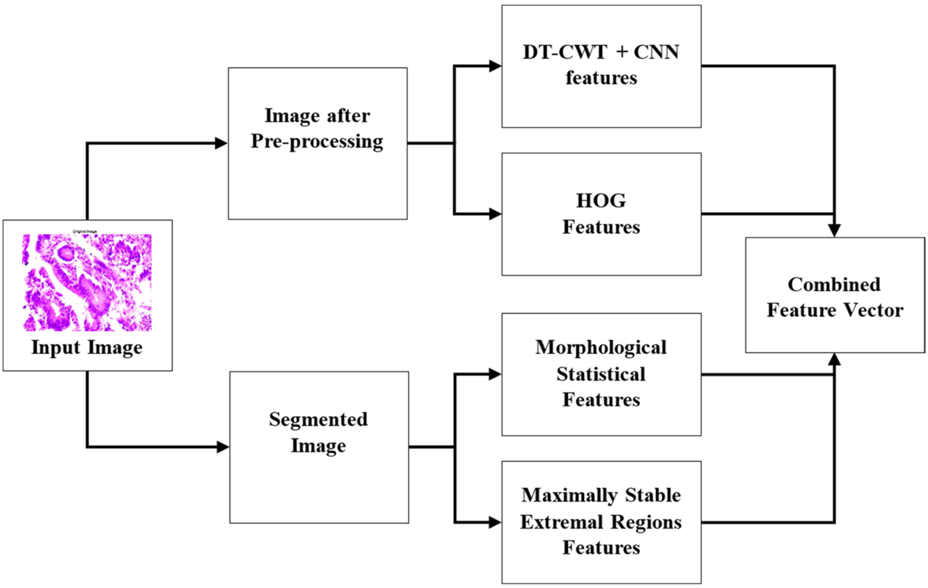


Fig. 5 Block diagram for the features extraction

- **Perimeter** $(\boldsymbol{P_i})$: The perimeter of R_i quantifies the boundary length of the segmented region:

$$P_i = \sum\nolimits_{(x,y)\in \partial R_i} 1 \tag{14}$$

Where ∂R_i indicates boundary pixels of R_i.

- **Circularity** $(\boldsymbol{C_i})$: This feature describes the shape of R_i by assessing its resemblance to a circle:

$$C_i = \frac{4\pi A_i}{P_i^2} \tag{15}$$

with values near 1 indicating a more circular shape, and lower values pointing to elongated or irregular forms.

- **Eccentricity** $(\boldsymbol{e_i})$: Eccentricity characterizes the elongation of the segmented region, calculated as the ratio of the distance between the foci of the ellipse that best fits R_i to its major axis length. Eccentricity values closer to 1 suggest elongated shapes.
- **Solidity** $(\boldsymbol{S_i})$: Solidity measures the proportion of the region's area to the area of its convex hull, reflecting the compactness of R_i:

$$S_i = \frac{A_i}{\text{ConvexHull}\,(A_i)} \tag{16}$$

4.4.2 Maximally Stable Extremal Regions (MSER) Features

MSER features are extracted from S_{hybrid} to identify stable regions that remain consistent over intensity variations, a characteristic particularly useful for emphasizing prominent cellular structures in colon tissue.

In this process, each region $R_i \subset S_{hybrid}$ undergoes a stability analysis across varying intensity thresholds. For a given intensity level t, an extremal region $E_{i,t}$ is defined where pixels either consistently exceed or fall below t. MSER extraction follows these steps [22]:

- **Intensity Stability**: An extremal region $E_{i,t}$ is identified by ensuring its pixel count remains stable over small variations in intensity t. The area consistency over intensity range t to $t + \delta\, t$ is given as:

$$\text{Stability}\,(E_{i,t}) = \frac{|A\,(E_{i,t+\delta\, t}) - A\,(E_{i,t})|}{A\,(E_{i,t})} \tag{17}$$

Where lower stability values indicate that $E_{i,t}$ is maximally stable.

- **Area Consistency**: Regions with low stability values are selected as MSERs, effectively capturing tissue structures that are unaffected by minor intensity changes, enhancing robustness to noise.

When MSER is used to find blobs in images, it may have little success with high-variation biopsy samples because of challenges posed by changes in lighting, staining, and unusual cell patterns. To solve the troubles that could arise due to missed or false region detection, the framework sets out to use various techniques. Initially, the colors in each image are adjusted to make them brighter and more normal, which helps MSER detection by clipping the effects of different stains. Second, we use MSER to form morphological descriptions and also combine them with HOG, DTCWT, and features obtained from a deep CNN. This means that details in missed regions can be found using different kinds of features. Thirdly, the areas found with MSER are checked against masks generated by the Mask R-CNN and Holistically-Nested Edge Detection (HED) networks. Once the regions matched with meaningful segments have been chosen, the remaining data contains fewer false positives. Also, ESO is used to pick out useful features, so the method rejects unreliable and noisy features that come from incorrect MSER regions. In this way, the main drawbacks of MSER are overcome, resulting in reliable and accurate detection of regions in images of colon biopsies.

4.4.3 DT-CWT + CNN features

The integration of DT-CWT with CNNs serves to leverage both frequency domain information and hierarchical spatial features, thereby providing a rich representation of the preprocessed image I'_n. This section delineates the processes involved in extracting features using DT-CWT and CNNs, culminating in a feature vector that encapsulates both sets of information.

DT-CWT decomposition The DT-CWT is applied to I'_n to obtain directional sub-bands at multiple scales. Each sub-band $W_{s,o}$ at scale s and orientation o is calculated as [23].

$$W_{s,o}\left(x,y\right)=\sum\nolimits_{u,v}I'_n\left(x-u,y-v\right)\psi_{s,o}\left(u,v\right) \tag{18}$$

Where $\psi_{s,o}$ represents the complex wavelet at s and o, and the sub-bands capture texture details at varying orientations and scales.

CNN feature extraction Following DT-CWT decomposition, the obtained wavelet coefficients are passed to a CNN to extract hierarchical features. The feature extraction process in CNNs typically involves several layers, including convolutional, activation, and pooling layers. The steps can be mathematically articulated as follows [24].

- **Convolutional Layer**: Each wavelet coefficient map $W_{s,o}\left(x,y\right)$ is convolved with a set of learnable filters K_j:

$$F_j\left(x,y\right)=\sum\nolimits_{m,n}W_{s,o}(x-m,y-n)\cdot K_j\left(m,n\right) \tag{19}$$

Where $F_j\left(x,y\right)$ denotes the feature map generated by the j^{th} filter. This operation extracts spatial features by identifying patterns within the wavelet-transformed images.

- **Activation Function**: An activation function, such as the Rectified Linear Unit (ReLU), is applied to introduce non-linearity:

$$A_j\left(x,y\right)=\max\left(0,F_j\left(x,y\right)\right) \tag{20}$$

- **Pooling Layer**: A pooling operation, typically Max Pooling, is performed to down-sample the feature maps, retaining only the most significant features:

$$P_j\left(x',y'\right)=\max_{(x,y)\in\, pooling-region}A_j\left(x,y\right) \tag{21}$$

Where $P_j\left(x',y'\right)$ represents the down-sampled feature map after pooling.

- **Fully Connected Layer**: After several convolutional and pooling layers, the resulting feature maps are flattened into a single vector f_{CNN} and passed through one or more fully connected layers:

$$f_{CNN}=W_f\cdot\ \text{flatten}\left(P_j\right)+b_f \tag{22}$$

Where W_f is the weight matrix and b_f is the bias vector. The final layer outputs a feature vector representing the extracted features from the CNN.

Fusion of DT-CWT and CNN features The features obtained from the DT-CWT f_{DT-CWT} and the CNN f_{CNN} are subsequently combined to create a comprehensive feature representation. This fusion process can be performed through concatenation.

$$f_{DT-CWT}=Flatten\left(DT-CWT\left(I_n'\right)\right) \tag{23}$$

$$f_{DT-CWT+CNN}=\left[f_{DT-CWT},f_{CNN}\right] \tag{24}$$

4.4.4 Histogram of oriented gradients (HOG) features

HOG features from the preprocessed image I_n' capture the distribution of gradient orientations within the tissue structures. HOG is instrumental in describing edge distributions and local gradient patterns, which aid in distinguishing structural details in colon tissue.

- ***Gradient Computation***: For each pixel (x,y) in I_n', gradients along the x and y-axes are computed as $G_x\left(x,y\right)$ and $G_y\left(x,y\right)$ [25]:

$$G_x\left(x,y\right)=\ I_n'\left(x+1,y\right)-I_n'\left(x-1,y\right) \tag{25}$$

$$G_y(x,y)=\ I_n'(x,y+1)-I_n'(x,y-1) \tag{26}$$

- ***Orientation and Magnitude Calculation***: The gradient magnitude $M\ (x, y)$ and orientation $\theta\ (x, y)$ at each pixel are determined as [25]:

$$M(x, y) = \sqrt{G_x(x, y)^2 + G_y(x, y)^2} \tag{27}$$

$$\theta\ (x, y) = \arctan\left(\frac{G_y(x, y)}{G_x(x, y)}\right) \tag{28}$$

- ***Histogram Construction***: The image is divided into small cells, and within each cell, the orientations are binned into histograms weighted by the gradient magnitude $M(x, y)$. This orientation histogram represents the structural edge information for each cell.
- ***Normalization***: Cells are grouped into blocks for contrast normalization, generating a descriptor vector that represents localized gradients in the image, sensitive to shape and edge distribution in tissue regions.

4.4.5 Combined feature vector

The combined feature vector $F_{combined}$ is created by concatenating all extracted features from previous sections, forming a comprehensive descriptor set. Each extracted feature type contributes a unique aspect of tissue characterization, from morphological details to texture patterns.

$$F_{combined} = [f_{Morphological}, f_{MSER}, f_{DT-CWT+CNN}, f_{HOG}] \tag{29}$$

This vector serves as the input to the classification module, effectively capturing the varied aspects of the tissue morphology and texture required for distinguishing pathological conditions in colon tissue. The construction of $F_{combined}$ ensures a balanced representation of shape, intensity, texture, and edge features.

4.5 Feature selection using enhanced snake optimization

This research work uses the ESO algorithm to refine the feature set derived from the combined feature vector $F_{combined}$. This optimization process aims to identify the most relevant features for colon cancer detection while minimizing redundancy among the selected features.

The combined feature vector is defined as:

$$F_{combined} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ . \\ . \\ . \\ f_m \end{bmatrix} \tag{30}$$

Where f_i represents individual features extracted from the image processing steps.

Energy function The ESO algorithm aims to minimize an energy function that encapsulates the effectiveness and efficiency of the selected feature subset. This energy function, E, can be expressed as [18].

$$E = E_{data} + E_{reg} + E_{red} \tag{31}$$

- ***Data Term*** $\boldsymbol{E_{data}}$: This term measures the discriminative ability of the features in classifying the data. It is calculated based on the performance of the selected features on the classification task:

$$E_{data} = \sum {}_{i=1}^{N} \sum {}_{j=1}^{C} \left(P_{ij} - \widehat{P}_{ij} \right)^2 \tag{32}$$

  Here, P_{ij} is the probability of the j^{th} class for the i^{th} sample derived from the selected features in $F_{combined}$.

- ***Regularization Term*** $\boldsymbol{E_{reg}}$: This term penalizes the selection of overly complex feature subsets, promoting simpler models. It is defined as:

$$E_{reg} = \lambda \sum {}_{k=1}^{m} w_k^2 \tag{33}$$

  Where w_k is the weight associated with the k^{th} feature in the selection process, and λ is a hyperparameter that balances the trade-off between accuracy and complexity.

- ***Redundancy Term*** $\boldsymbol{E_{red}}$: This term quantifies the redundancy among the features selected from $F_{combined}$ to ensure diversity:

$$E_{red} = \sum {}_{k=1}^{m} \sum {}_{l=k+1}^{m} \rho \left(w_k, w_l \right) \tag{34}$$

  Here, $\rho\ (w_k, w_l)$ is a metric that measures the correlation between features k and l, ensuring that the selected features provide complementary information.

  Optimization Steps:

- ***Initialization***: The ESO algorithm initializes a population of candidate feature subsets from $F_{combined}$.
- ***Evaluation***: Each candidate feature subset is evaluated using the defined energy function E. The quality of the subsets is assessed based on their corresponding energy values.
- ***Selection and Iteration***: The best-performing subsets are retained, and the algorithm iteratively refines the selection through mutation and crossover operations, enhancing the diversity of candidate solutions.
- ***Final Output***: The output of the ESO process is the selected feature subset denoted as $F_{selected}$, which contains the most relevant features from $F_{combined}$ necessary for effective colon cancer classification.

As a result of this, the number of important features used for analysis and classification stays the same, even though the rest are removed. The operation of ESO for feature selection is shown in Fig. 6.

Using the ESO algorithm allows choosing the most important and unique features from the hybrid data, leading to both a better classification result and faster computations.

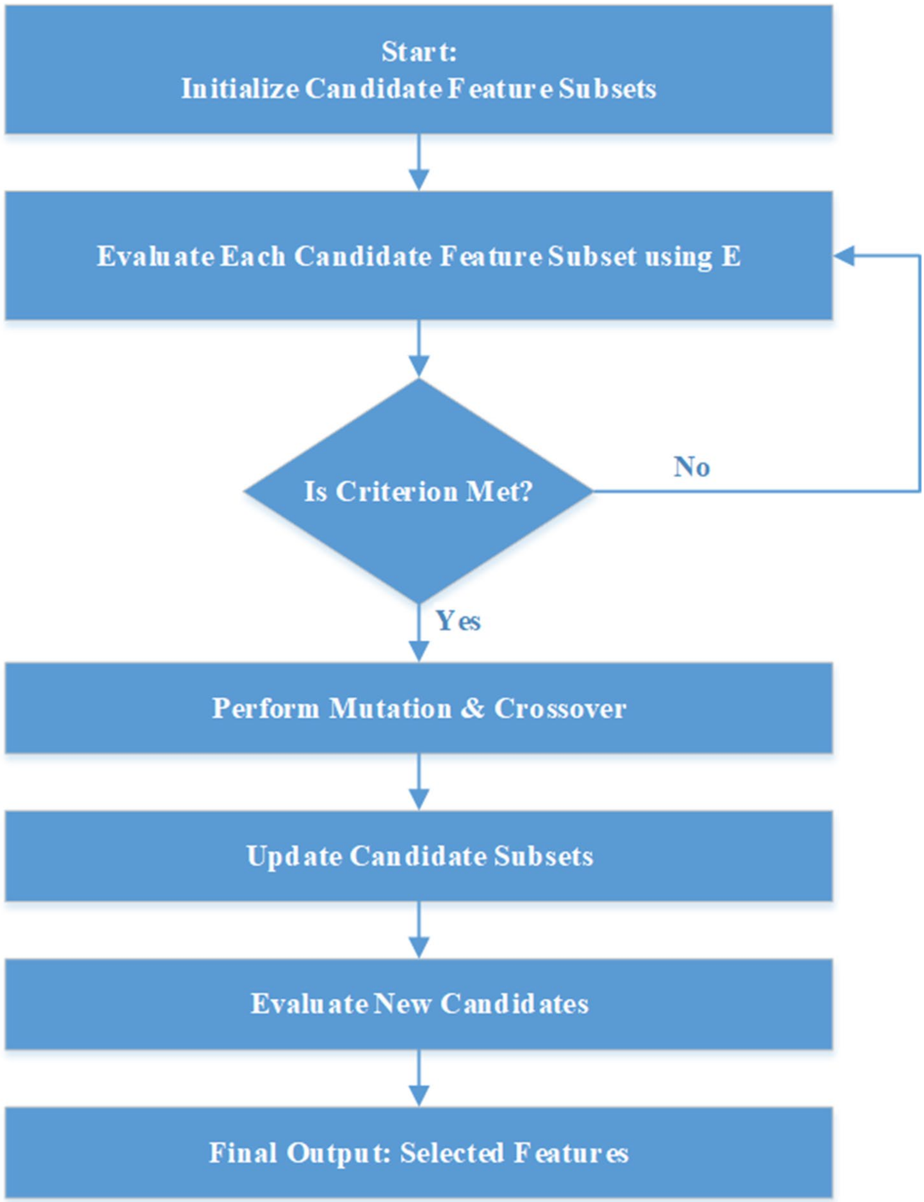


Fig. 6 Flow Diagram of Enhanced Snake Optimization for Feature Selection

4.6 Classification using Dual-Classifier framework

In this section, a dual-classifier framework is presented that effectively classifies the selected feature set $F_{selected}$. The approach integrates two distinct classifiers: Long Short-Term Memory (LSTM) networks and Random Forest (RF) classifiers. This combination leverages the strengths of both models to enhance classification performance in colon cancer detection.

4.6.1 Long Short-Term memory (LSTM) classifier

LSTM networks are a type of recurrent neural network (RNN) specifically designed to learn from sequences of data [26]. They are particularly adept at capturing temporal dependencies and can manage long-range correlations in input data. Given the sequential nature of image features derived from $F_{selected}$, LSTM networks are well-suited for our classification task.

Mathematical representation The input to the LSTM network at time step t is represented as [26].

$$x_t = F_{selected}\left[t\right] \tag{35}$$

The LSTM cell utilizes three gates to control the information flow: the input gate i_t , the forget gate f_t, and the output gate o_t. The operations can be summarized as follows [26]:

- ***Forget Gate***:

$$f_t = \sigma\left(W_f \cdot \left[h_{t-1}, x_t\right] + b_f\right) \tag{36}$$

- ***Input Gate***:

$$i_t = \sigma\left(W_i \cdot \left[h_{t-1}, x_t\right] + b_i\right) \tag{37}$$

$$\widetilde{C}_t = \tanh\left(W_C \cdot \left[h_{t-1}, x_t\right] + b_C\right) \tag{38}$$

- ***Cell State Update***:

$$C_t = f_t \odot C_{t-1} + i_t \odot \widetilde{C}_t \tag{39}$$

- ***Output Gate***:

$$o_t = \sigma\left(W_o \cdot \left[h_{t-1}, x_t\right] + b_o\right) \tag{40}$$

$$h_t = o_t \odot \tanh\left(C_t\right) \tag{41}$$

Finally, the classification output is produced by passing the last hidden state through a softmax activation function:

$$y_t = \text{softmax}\left(W_y \cdot\ h_t + b_y\right) \tag{42}$$

Where W_y and b_y are the output layer weights and biases, respectively.

4.6.2 Random forest (RF) classifier

The Random Forest classifier operates by constructing multiple decision trees during training and outputting the mode of their predictions. This ensemble method enhances robustness and mitigates overfitting, making it particularly effective for high-dimensional feature spaces like $F_{selected}$.

Mathematical representation For a given input feature vector $F_{selected}$, the Random Forest classifier generates predictions from N decision trees [27].

$$y_{RF} = \frac{1}{N}\sum{}_{i=1}^{N}T_i\left(F_{selected}\right) \tag{43}$$

Where T_i denotes the prediction from the i^{th} tree. The final prediction is made by selecting the class with the highest aggregated score across all trees.

4.6.3 Integration of classifiers

To leverage the strengths of both classifiers, we propose an ensemble strategy that combines the outputs of the LSTM and RF classifiers. The final decision is made through majority voting:

- ***Predictions***: Let y_{LSTM} be the output from the LSTM classifier and y_{RF} from the Random Forest.

A strategy for optimization using validation is used to support the LSTM and Random Forest classifiers in the proposed dual-classifier framework. The process relies on several topics that are mathematically based.

1. Cross-Validation Setup.

Let the data be shown as $=\{(x_i, y_i)\}^N{}_{i=1}$, where x_i is the information contained in the data and y_i represents the actual outcome for that information. To carry out k-fold cross-validation, the data is divided into subsets $D_1, D_2, \ldots\ \ldots\ .D_k$.

For $j \in\ \{1, \ldots\ ., k\}$ training and validation sets are given as:

$$D_{train}^{(j)} = D\backslash\ D_j,\ D_{Val}^{(j)} = D_j \tag{44}$$

Each fold is checked using the performance metrics mentioned below:

- Accuracy:

$$Acc^{(j)} = \frac{1}{|D_j|} \sum x, y \in \ D_j \in \prod \mid \widehat{y} = y| \tag{45}$$

- F1-Score: It is the average of the precision and recall per fold.

2. Parameter Tuning.

Let:

- $\Theta_{LSTM} = \{\eta, h, e\}$ and η stands for the learning rate, h is the number of hidden units and e is the number of epochs.
- Θ_{RF} consists of T trees and d, the maximum depth.

The hyperparameters are tuned by finding the combination with the greatest average F1-score among all the k folds.

$$\Theta^{*} = argmax_{\Theta} \frac{1}{k} \sum {}_{j=1}^{k} F_1^{(j)} (\Theta) \tag{46}$$

3. Tie-Breaking Preference Validation.

An ensemble classification tie appears if the predictions of the LSTM and RF are not the same for the same input, i.e. $\widehat{y}LSTM(x) \neq \widehat{y}RF(x)$

Let:

- C_{LSTM} represents the accuracy of classification when using an LSTM alone.
- C_{RF} symbolizes accuracy when only using RF.

Define the tie-breaking rule as:

$$\widehat{y}\,Final(x) = \begin{cases} \widehat{y}\,LSTM(x), \ if\, \widehat{y}\,LSTM(x) = \widehat{y}\,RF(x) \\ \widehat{y}\,LSTM(x), \ if\, \widehat{y}\,LSTM(x) \neq \widehat{y}\,RF(x) \\ \widehat{y}RF(x) \ \ otherwise \end{cases} \bigwedge C_{LSTM} \succ C_{RF} \tag{47}$$

The rule ensures that the classifier with better accuracy during validation is used in such situations.

This framework enables the model to make informed classification decisions, utilizing the temporal and spatial features captured by LSTM and the robust decision-making capabilities of the Random Forest classifier. The dual-classifier setup aims to improve the accuracy and reliability of classifications in colon cancer detection, making it a valuable contribution to the field.

4.6.4 Pseudo code for dual classifier

```
Input: F_selected (feature set)
Output: y_final (final classification output)
Initialize LSTM model
Initialize Random Forest model
Train LSTM model using F_selected
Train Random Forest model using F_selected
Initialize an empty list for predictions: LSTM_predictions, RF_predictions
FOR each feature vector f in F_selected DO
   LSTM_output ← LSTM_model.predict(f)
   RF_output ← RF_model.predict(f)
   Append LSTM_output to LSTM_predictions
   Append RF_output to RF_predictions
END FOR
Initialize an empty list for final predictions: final_predictions
FOR i = 1 to length(LSTM_predictions) DO
   lstm_pred ← LSTM_predictions[i]
   rf_pred ← RF_predictions[i]
   IF lstm_pred == rf_pred THEN
   final_predictions.append(lstm_pred) // No conflict
ELSE
   // Tie-breaking strategy
   final_predictions.append(lstm_pred) // Prefer LSTM prediction
   // Alternative strategies can be implemented
   // final_predictions.append(rf_pred) // Prefer RF prediction
   // final_predictions.append(random_choice(lstm_pred, rf_pred)) // Random choice
 END IF
END FOR
RETURN final_predictions
```

5 Results and discussion

5.1 Dataset description

The framework under consideration is assessed using colon pathological image datasets sourced from four distinct origins: two from Indian hospitals and two from established benchmark datasets. These datasets encompass various locations and microscopy magnifications, representing normal and malignant categories (well-differentiated, moderately differentiated, and poorly differentiated). Table 2 summarizes key information about the datasets used in the proposed framework, including the number of images, resolution, tissue thickness, magnifications, and the entities involved in the analysis. Additionally, Table 2 presents representative images from the different colon pathological datasets, illustrating the variations in appearance across normal and malignant categories.

- Ishita Pathology Center (IPC) dataset [19]: This dataset comprises 1200 images obtained from H&E-stained colon biopsy samples at various magnifications (4X, 10X, and 40X). The images were captured from tissue sections with a thickness of 5–6 µm using a Magcam CD5 microscope coupled with an Olympus CX33. Dr. Ranjana Srivastava analyzed the slides and generated the corresponding ground truth labels.
- Aster Medcity (AMC) dataset [19]: Featuring 840 images at a resolution of 640 × 480, this dataset includes samples from H&E-stained colon biopsies. The images were captured at magnifications of 10X, 20X, and 40X. The analysis was performed by Dr. Sarah Kuruvila and Dr. Shahin Hameed, who provided the ground truth labels for the dataset.

Table 2 Overview of colon pathological image datasets

Dataset Name	Number of Images	Resolution	Tissue Thickness	Magnifications	Normal	Well	Moderate	Poor	Analysis Conducted By
Ishita Pathology Center (IPC) [1]	1200	640 × 480	5–6 µm	4X, 10X, 40X	100	100	100	100	Dr. Ranjana Srivastava, IPC, Allahabad, India
Aster Medcity (AMC) [19]	840	640 × 480	5–6 µm	10X, 20X, 40X	70	70	70	70	Dr. Sarah Kuruvila, Aster Medcity, Kochi, India; Dr. Shahin Hameed, MVR Cancer Center, Kerala
GlaS Dataset [28]	165	640 × 480	4–5 µm	20X	74	-	47	24	Korsuk Sirinukunwattana, David R. J. Snead, and Nasir M. Rajpoot
Imediatreat Dataset [29]	357	800 × 600	5–6 µm	10X	62	96	99	100	Catalin Stoean, Ruxandra Stoean, Adrian Sandita, Daniela Ciobanu, Cristian Mesina and Corina Lavinia Gruia

- GlaS dataset [28]: Consisting of 165 images at a 20X magnification, this dataset has been labeled by an expert pathologist, categorizing them into normal and varying degrees of malignancy (moderate and poor).
- Imediatreat dataset [29]: This collection includes 357 images obtained at 10X magnification. The dataset categorizes the images based on malignancy grades, ranging from normal to third grade (Table 3).

5.2 Simulation parameters

Table 4 presents various simulation parameters used in this research work.

5.3 Evaluation parameters

Table 5.

Other evaluation parameters for the proposed approach:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{48}$$

$$Precision = \frac{TP}{TP + FP} \tag{49}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{50}$$

$$Specificity = \frac{TN}{TN + FN} \tag{51}$$

$$Error\ Rate = \frac{FP + FN}{TP + TN + FP + FN} \tag{52}$$

$$False\ Positive\ Rate\ (FPR) = \frac{FP}{FP + TN} \tag{53}$$

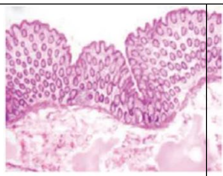
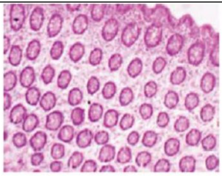
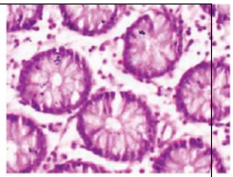
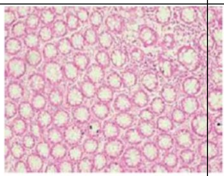
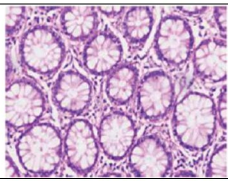
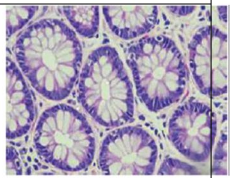
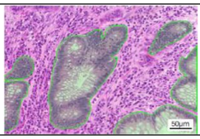
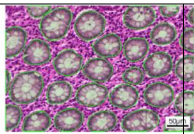
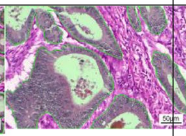
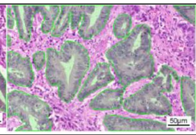
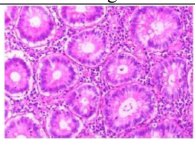
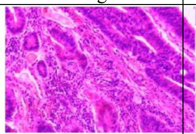
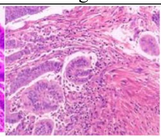
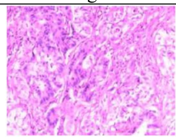
$$F - Score = \frac{2TP}{2TP + FP + FN} \tag{54}$$

5.4 Results

Table 6

From Table 7, it can be seen that for image enhancement parameters, Edge Content (EC) shows the highest value in Dataset2 (3.45), suggesting sharper edges compared to the other datasets, while Dataset4 has the lowest EC (1.89), potentially indicating less pronounced edge features. Enhancement Measure of Emphasis (EME) is also highest in Dataset2 (2.03), indicating improved image contrast and detail, whereas Dataset4 has the lowest EME (0.986), suggesting minimal enhancement effects.

Table 3 Sample images from various datasets

IPC Dataset [19]				
	4X	10X		40X
AMC Dataset [19]				
	4X	10X		40X
GlaS Dataset [28]				
	Benign	Benign	Malignant	Malignant
Imediatre at dataset [29]				
	Normal (Grade 0)	Cancer Grade 1 (G1)	Cancer Grade 2 (G2)	Cancer Grade 3 (G3)

The image segmentation parameters provide insight into segmentation quality across datasets. Normalized Cross Correlation (NCC) ranges from 0.816 in Dataset2 to 0.932 in Dataset4, with Dataset4 showing the highest NCC, indicating better alignment and similarity to the reference. Mutual Information (MI), which measures shared information between images, peaks at 7.94 in Dataset4, reflecting a higher degree of feature alignment and information sharing, especially compared to Dataset2, where MI is lowest (4.394). Boundary Displacement Error (BDE), an indicator of boundary alignment accuracy, is lowest in Dataset4 (4.03), implying a more precise boundary match in segmentation, while Dataset1 has the highest BDE (4.98), suggesting less accurate boundary alignment. Feature Similarity Index (FSIM) scores are relatively close across datasets, with Dataset4 scoring highest (0.894), which indicates high structural and feature similarity in segmented images. Overall, Dataset4 demonstrates strong performance across most metrics, showing better enhancement and segmentation quality. Dataset2 shows higher edge contrast but lower segmentation quality, as indicated by lower NCC and MI values (Fig. 7).

Table 8 outlines the output of the colon biopsy images, which reveals that the model identified the normal and malignant tissues correctly. This also shows that malignant samples are assigned as well, moderately, or poorly differentiated and helps illustrate the differences in their physical characteristics.

Tables 9 and 10, and 11 provide a summary of the performance analysis of colon cancer detection using the AMC dataset [19] at various magnifications (10X, 20X, and 40X). The

Table 4 Simulation parameters for LSTM, RF, and ESO

LSTM Parameters	Values
Input Dimensions	512
Hidden Layer Size	128
Number of LSTM Layers	2
Dropout Rate	0.3
Learning Rate	0.001
Batch Size	32
Total Epochs	100
Random Forest (RF) Parameters	**Values**
Number of Trees	200
Maximum Depth	15
Minimum Samples Split	4
Minimum Samples Leaf	2
Criterion	Gini
Enhanced Snake Optimization (ESO) Parameters	**Values**
Population Size	50
Maximum Iterations	150
Inertia Weight (ω)	0.6
Attraction Coefficient (α)	1.5
Repulsion Coefficient (β)	1.3
Initial Position Range	[−1, 1]
Velocity Range	[−0.1, 0.1]

detection accuracy at 10X magnification is shown in Table 9, where Hybrid Features had the best accuracy (92.50%), closely followed by DT–CWT+CNN (91.09%). Table 10's performance metrics at 20X magnification demonstrate that Hybrid Features' accuracy increased to 93.70%, demonstrating the advantages of more detailed image analysis, while the Morphological method's accuracy was 91.40%. The findings at 40X magnification are finally shown in Table 11, where Hybrid Features further increased to 94.80%, proving the effectiveness of sophisticated feature extraction methods at higher magnifications. These findings show a distinct tendency for better performance with higher magnification, highlighting the significance of accurate image processing in the identification of colon cancer.

The performance study of colon cancer detection using the IPC dataset [19] at 4X, 10X, and 40X magnifications is shown in Tables 12 and 13, and 14. The morphological characteristics in Table 12 recorded an accuracy of 88.50%, while the hybrid features obtained the maximum accuracy of 90.40%. In Table 13, the accuracy of hybrid characteristics improved to 92.00% and morphological features to 90.00% when the magnification was increased to 10X. The hybrid features achieved the maximum accuracy of 92.20%, while the morphological characteristics reached 91.20%, according to Table 14, which displays the findings of the 40X magnification. The findings show that all feature sets benefit from increased magnification in terms of colon cancer diagnosis accuracy, with hybrid features continuously beating the others.

The performance study of colon cancer detection with the GlaS dataset [28] at a 20X magnification is shown in Table 15. The hybrid features demonstrated the efficacy of integrated feature extraction strategies by achieving the greatest accuracy of 95.10%. Additionally, the DT–CWT+CNN features performed well, achieving 93.50% accuracy. Accuracy values for morphological and HOG characteristics were 91.50% and 92.80%, respectively.

Table 5 Evaluation parameters for image preprocessing and image segmentation

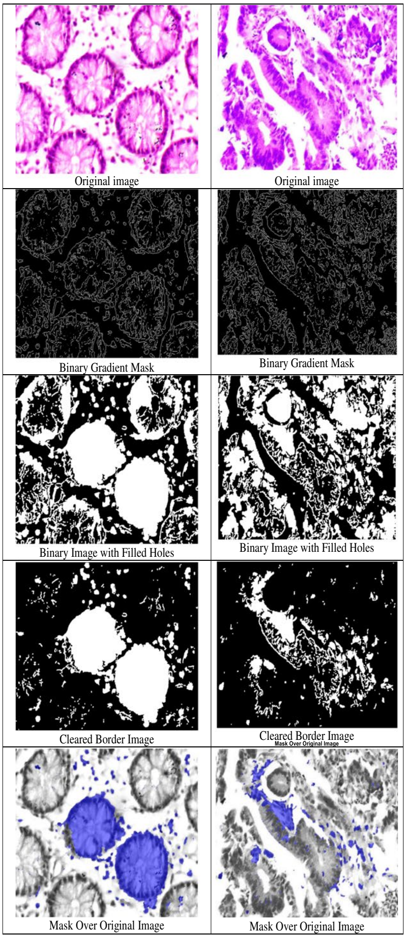
Parameters	Descriptions	Formulation
Image Preprocessing Parameters		
EC (Edge Content)	Evaluates contrast enhancement based on image gradients.	$EC = \frac{1}{M\times N}\sum_{x}\sum_{y}\lvert G(x,y)\rvert$ Where M and N are image dimensions, and $G(x,y)$ is the image gradient.
EME (Enhancement Measure of Emphasis)	Measures the degree of contrast enhancement.	$EME = \frac{1}{b_1 b_2}\sum_{j=1}^{b_2}\sum_{i=1}^{b_1} 20\log\frac{I_{max}}{I_{min}}$
Image Segmentation Parameters		
NCC (Normalized Cross Correlation)	Measures the similarity between segmented and ground truth images.	$NCC = \frac{\sum\left(I_s(x,y)-\bar{I}_s\right)\left(I_g(x,y)-\bar{I}_g\right)}{\sqrt{\sum\left(I_s(x,y)-\bar{I}_s\right)^2\sum\left(I_g(x,y)-\bar{I}_g\right)^2}}$ Where $I_s(x,y)$ and $I_g(x,y)$ are the segmented and ground truth images, and $\bar{I}_s$, $\bar{I}_g$ are their means.
MI (Mutual Information)	Evaluates the mutual dependence between segmented and ground truth images.	$MI = \sum P(I_s,I_g)\log\left(\frac{P(I_s,I_g)}{P(I_s)P(I_g)}\right)$ Where $P(I_s,I_g)$ is the joint probability distribution of I_s and I_g, and $P(I_s)$, $P(I_g)$ are their marginal distributions.
BDE (Boundary Displacement Error)	Measures average distance between boundaries of segmented and ground truth images.	$BDE = \frac{1}{N}\sum_{p\in B_s}\min_{q\in B_g}\mathrm{dist}(p,q)$ Where B_s and B_g represent boundaries in the segmented and ground truth images, and $\mathrm{dist}(p,q)$ is the Euclidean distance between points p and q.
FSIM (Feature Similarity Index)	Assesses perceptual similarity based on image features.	$FSIM = \frac{\sum_{x,y}S_{PC}(x,y)\cdot S_G(x,y)\cdot L(x,y)}{\sum_{x,y}S_{PC}(x,y)}$ Where $S_{PC}(x,y)$ and $S_G(x,y)$ represent phase congruency and gradient similarity, and $L(x,y)$ denotes luminance similarity.

The findings highlight the GlaS dataset's dependability and efficacy in colon cancer detection tasks by showing that it regularly produces higher accuracy across all feature sets when compared to earlier datasets. GlaS is known as a benchmark dataset.

The performance study of detecting colon cancer using the Imediatreat dataset [29] at a 10X magnification is shown in Table 16. With the best accuracy of 98.70%, the hybrid features demonstrated the efficacy of integrated feature extraction methods in this benchmark dataset. Other feature sets, such as HOG and DT–CWT+CNN, also showed remarkable accuracy rates of 98.30% and 98.50%, respectively. The findings demonstrate the Imediatreat dataset's outstanding performance, demonstrating its dependability and efficacy for colon cancer diagnosis, surpassing earlier datasets in terms of accuracy and precision across all assessed criteria.

The proposed model shows similar performance during various magnification scales (4X, 10X, 40X). The model was tested in every magnification state, and the performance shows that it is reliable at every scale.

Table 6 Segmentation results for two different input images



- When using 4X Magnification, the model presented an accuracy rate of 90.40%, of which there was a precision rate of 89.00% and a recall rate of 88.10%. Such values show that the model is effective in the recognition of wider constructs of tissues at lower magnification frequencies.
- At 10X Magnification, the accuracy changed to 92.00%, precision to 90.30% and recall to 89.50% which indicated that the model is even more specifically contributing to detecting and segmenting tissue groups with a higher level of clarity at 10X magnification.
- With the model on 40X Magnification, its accuracy was increased to 92.20%, precision was 91.80%, and recall was at 90.90%, and the detection of smaller cellular features was enhanced, which matters at high magnifications.

These findings assure that the model performs quite well at all magnifications. The small increases in performance with higher degrees of magnification can be explained by the finer granularity of the features captured in higher magnifications, whereas the overall lack

Table 7 Outcome of image enhancement and image segmentation on all the dataset

Parameters Image Enhancement	DATAset1	Dataset2	Dataset3	Dataset4
EC (Edge Content)	2.56	3.45	2.58	1.89
EME (Enhancement Measure of Emphasis)	1.045	2.03	1.034	0.986
Image Segmentation Parameters				
NCC (Normalized Cross Correlation)	0.907	0.816	0.9234	0.932
MI (Mutual Information)	5.032	4.394	4.848	7.94
BDE (Boundary Displacement Error)	4.980	4.30	4.10	4.03
FSIM (Feature Similarity Index)	0.8729	0.815	0.869	0.894

of variation in precision and recall among magnification degrees supports the argument of magnification invariance.

5.5 Statistical significance analysis

To measure the improvements accurately, we used statistics to compare the Hybrid Features approach to Morphological, MSER, DT–CWT + CNN and HOG individually. Tests such as

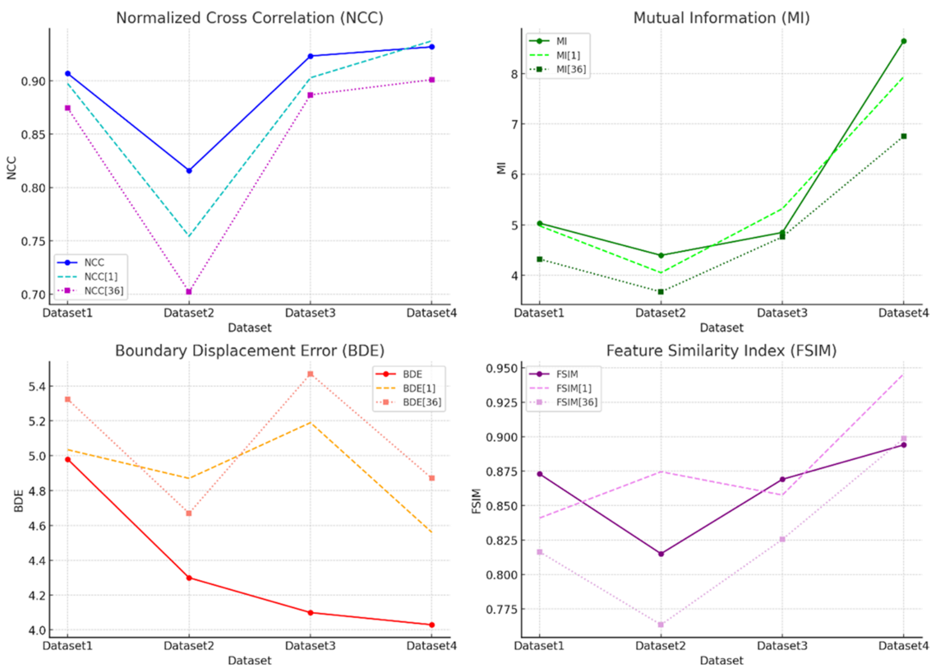
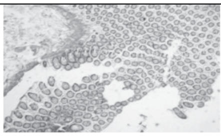
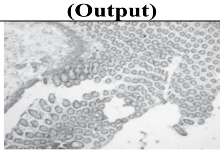
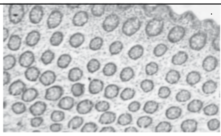
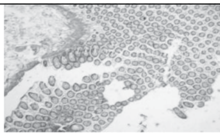
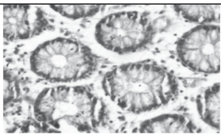
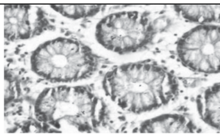
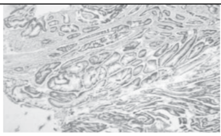
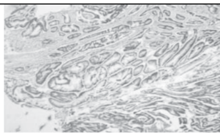
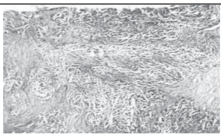
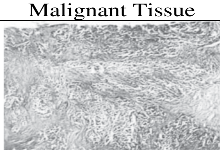


Fig. 7 comparative analysis of the image segmentation outcome with state of arts

Table 8 Classified output of colon biopsy images based on detected tissue type and grade

S. No.	Input Biopsy Image	Detected Classification (Output)	Tissue Grade
1		 Normal Tissue	Normal
2		 Normal Tissue	Normal
3		 Normal Tissue	Normal
4		 Malignant Tissue	Well Differentiated (Grade I)
5		 Malignant Tissue	Moderately Differentiated (Grade II)
6		 Malignant Tissue	Poorly Differentiated (Grade III)

the paired t-test and Wilcoxon signed-rank test were carried out using several main criteria (e.g., Accuracy, Sensitivity, Precision, etc.).

According to the results, using Hybrid Features provided statistically better outcomes when compared to all other methods, having p-values < 0.05 during both tests.

- Morphological vs. Hybrid: *t-test* $p = 0.0143$, $p = 0.0195$.
- MSER vs. Hybrid: *t-test* $p = 0.0129$, *Wilcoxon* $p = 0.0195$.
- DT–CWT + CNN vs. Hybrid: *t-test* $p = 0.0149$, *Wilcoxon* $p = 0.0195$.
- HOG vs. Hybrid: *t-test* $p = 0.0142$, *Wilcoxon* $p = 0.0195$.

This means that the improvements came from appropriate strategies and not by chance. The outcome further proves that our technique can accurately detect cancer from histopathological images.

Table 9 Performance analysis of colon cancer detection using 10X magnification for AMC dataset

Parameters	Morphological	MSER	DT–CWT + CNN	HOG	Hybrid Features
Accuracy	90.70%	90.00%	91.09%	90.23%	92.50%
Error Rate	9.30%	10.00%	8.91%	9.77%	7.50%
Sensitivity	90.40%	89.80%	91.20%	87.10%	92.30%
Specificity	96.90%	96.70%	96.90%	96.10%	97.50%
Precision	91.10%	90.80%	92.20%	91.50%	93.70%
False Positive Rate	3.10%	3.30%	3.10%	3.90%	2.50%
F1-Score	90.75%	89.80%	91.60%	88.50%	92.90%
MCC	0.873	0.870	0.889	0.865	0.910
Kappa Statistics	0.765	0.740	0.790	0.735	0.820

Table 10 Performance analysis of colon cancer detection using 20X magnification for AMC dataset

Parameters	Morphological	MSER	DT–CWT + CNN	HOG	Hybrid Features
Accuracy	91.40%	90.80%	92.50%	91.00%	93.70%
Error Rate	8.60%	9.20%	7.50%	9.00%	6.30%
Sensitivity	91.20%	90.50%	92.70%	88.80%	93.40%
Specificity	97.20%	96.90%	97.00%	96.50%	98.10%
Precision	92.00%	91.50%	93.40%	91.80%	94.60%
False Positive Rate	2.80%	3.10%	3.00%	3.50%	1.90%
F1-Score	91.70%	90.90%	92.90%	89.50%	93.90%
MCC	0.885	0.870	0.901	0.865	0.930
Kappa Statistics	0.798	0.762	0.810	0.740	0.860

Table 11 Performance analysis of colon cancer detection using 40X magnification for AMC dataset

Parameters	Morphological	MSER	DT–CWT + CNN	HOG	Hybrid Features
Accuracy	92.00%	91.50%	93.00%	91.80%	94.80%
Error Rate	8.00%	8.50%	7.00%	8.20%	5.20%
Sensitivity	92.50%	91.20%	93.40%	89.50%	94.50%
Specificity	97.50%	97.20%	97.30%	96.80%	98.30%
Precision	93.00%	92.20%	94.20%	92.50%	95.00%
False Positive Rate	3.50%	2.80%	2.70%	3.20%	1.70%
F1-Score	92.40%	91.70%	93.70%	90.00%	94.20%
MCC	0.902	0.885	0.915	0.880	0.940
Kappa Statistics	0.820	0.805	0.835	0.795	0.875

Table 12 Performance analysis of colon cancer detection using 4X magnification for IPC dataset

Parameters	Morphological	MSER	DT–CWT+CNN	HOG	Hybrid Features
Accuracy	88.50%	87.30%	89.20%	86.80%	90.40%
Error Rate	11.50%	12.70%	10.80%	13.20%	9.60%
Sensitivity	88.10%	86.80%	88.90%	85.60%	89.50%
Specificity	91.20%	90.00%	92.10%	89.70%	91.90%
Precision	89.00%	87.50%	90.80%	88.20%	90.30%
False Positive Rate	8.80%	10.00%	7.90%	11.30%	8.10%
F1-Score	88.40%	86.30%	89.50%	86.00%	89.80%
MCC	0.7900	0.7800	0.8100	0.7500	0.8000
Kappa Statistics	0.6700	0.6400	0.7000	0.6200	0.6800

Table 13 Performance analysis of colon cancer detection using 10X magnification for IPC dataset

Parameters	Morphological	MSER	DT–CWT+CNN	HOG	Hybrid Features
Accuracy	90.00%	89.50%	90.80%	89.00%	92.00%
Error Rate	10.00%	10.50%	9.20%	11.00%	8.00%
Sensitivity	89.50%	88.20%	90.30%	88.70%	90.50%
Specificity	93.00%	92.50%	93.50%	91.80%	93.20%
Precision	90.30%	89.20%	91.00%	89.50%	91.50%
False Positive Rate	7.00%	7.50%	6.50%	8.20%	6.80%
F1-Score	89.90%	88.70%	90.80%	88.20%	90.70%
MCC	0.8500	0.8300	0.8600	0.8100	0.8400
Kappa Statistics	0.7300	0.7100	0.7400	0.7000	0.7200

Table 14 Performance analysis of colon cancer detection using 40X magnification for IPC dataset

Parameters	Morphological	MSER	DT–CWT+CNN	HOG	Hybrid Features
Accuracy	91.20%	90.50%	91.00%	90.30%	92.20%
Error Rate	8.80%	9.50%	9.00%	9.70%	7.80%
Sensitivity	90.90%	89.80%	90.50%	89.30%	91.00%
Specificity	94.50%	93.70%	94.80%	92.90%	94.00%
Precision	91.80%	90.90%	91.30%	90.50%	91.60%
False Positive Rate	5.50%	6.00%	5.50%	7.10%	6.30%
F1-Score	90.80%	89.70%	90.20%	89.10%	91.40%
MCC	0.8600	0.8400	0.8500	0.8300	0.8700
Kappa Statistics	0.7500	0.7300	0.7400	0.7200	0.7600

Table 15 Performance analysis of colon cancer detection using 20X magnification for GlaS dataset

Parameters	Morphological	MSER	DT–CWT+CNN	HOG	Hybrid Features
Accuracy	91.50%	92.00%	93.50%	92.80%	95.10%
Error Rate	0.0850	0.0800	0.0650	0.0700	0.0500
Sensitivity	91.20%	92.50%	93.00%	92.10%	94.90%
Specificity	97.00%	97.20%	97.50%	97.10%	97.80%
Precision	92.50%	93.00%	94.50%	93.80%	95.00%
False Positive Rate	0.0300	0.0280	0.0250	0.0290	0.0200
F1-Score	91.70%	92.80%	93.80%	92.40%	95.00%
MCC	0.9100	0.9200	0.9300	0.9200	0.9500
Kappa Statistics	0.8200	0.8300	0.8400	0.8300	0.8600

Table 16 Performance analysis of colon cancer detection using 10X magnification for imediatreat dataset

Parameters	Morphological	MSER	DT–CWT+CNN	HOG	Hybrid Features
Accuracy	98.10%	98.20%	98.50%	98.30%	98.70%
Error Rate	0.0100	0.0080	0.0070	0.0075	0.0050
Sensitivity	98.00%	98.40%	98.60%	98.20%	98.80%
Specificity	99.00%	99.10%	99.20%	99.00%	99.50%
Precision	98.20%	98.50%	98.80%	98.60%	99.00%
False Positive Rate	0.0050	0.0040	0.0035	0.0045	0.0030
F1-Score	98.05%	98.20%	98.40%	98.30%	98.70%
MCC	0.9700	0.9750	0.9800	0.9750	0.9850
Kappa Statistics	0.9400	0.9450	0.9500	0.9450	0.9600

Table 17 shows the classification accuracy of the proposed framework for all four datasets, from Grade 0 to Grade 3 for each one. Naturally, the highest accuracy is recorded for Normal images, with results declining as malignancy increases, as the cells grow more complex and share similar characteristics. In general, Imediatreat returns the highest Likelihood of correct results for damaged tissue and Poorly Differentiated samples, reaching 98.70% and 92.80%, respectively. In this case, with GlaS, the lower scale values reflect the variable nature and difficult parts in the images. Such an analysis solves the reviewer's concern and ensures that the proposed approach is dependable when ranking colon biopsy images from various datasets.

Table 18 provides information about different ways to find colon cancer and highlights results for precision, recall (sensitivity), F1-score, and accuracy. Hamida et al. [13] applied a CNN model that achieved an accuracy of 96.98%, but more details on the other metrics were not given. In addition, although Kavitha et al. [14] used a VGG-CNN, their lack of precision and F1-score information means we cannot review the model in full detail. Vasu K [15]. applied a simple ResNet-18 classifier, leading to an accuracy of 86.50% on the

Table 17 Grade-wise accuracy for all datasets (AMC, IPC, glas, Imediatreat)

Dataset	Grade 0 (Normal)	Grade 1 (Well Differentiated)	Grade 2 (Moderately Differentiated)	Grade 3 (Poorly Differentiated)
AMC	98.70%	96.80%	94.10%	91.30%
IPC	97.10%	94.90%	92.20%	89.70%
GlaS	95.50%	93.30%	91.00%	88.20%
Imediatreat	98.70%	97.10%	95.30%	92.80%

Table 18 Performance analysis of proposed colon cancer detection with previous research works

Method		Precision	Recall/ Sensitivity	F1-Score	Accuracy
Hamida et al. [13]	CNN	--	--	--	96.98%
Kavitha et al. [14]	VGG-CNN	--	95.10%	--	93.48%
Vasu, K. [15]	Resnet-18 classifier	--	--	--	86.50%
Karthikeyan et al. [30]	ANN	87%	89%	--	88%
	BPNN	91%	91%	--	92%
	Ranking Algorithm with CNN	92%	93%	--	91%
Suganya et al. [31]	SVM+RF	--	--	--	72.1%
	GLCM+SVM	--	--	--	81.7%
Babu et al. [19]	2DReCA/hybrid features/ RF	92.93	100%	96.30%	96.48%
Proposed Hybrid Features-based Approach with AMC dataset		**92.50%**	**98.80%**	**98.70%**	**98.70%**
Proposed Hybrid Features-based Approach with IPC dataset		**91.09%**	**97.80%**	**97.10%**	**97.10%**
Proposed Hybrid Features-based Approach with GlaS dataset		**95.00%**	**96.00%**	**95.50%**	**95.50%**
Proposed Hybrid Features-based Approach with Imediatreat dataset		**98.00%**	**98.80%**	**98.70%**	**98.70%**

dataset. Karthikeyan et al. [30] focused on applying three unique methods. Developing the ANN model allowed them to attain 88% accuracy, 87% precision, and 89% recall. This form of the neural network achieved almost perfect accuracy and almost the same friendly precision and recall of 91%. Ranking Algorithm with CNN achieved precision and recall values of 92% and 93% and an accuracy of 91%, yet the F1-scores were not included in the report. According to Suganya et al. [31], the SVM paired with RF reached 72.1% accuracy, but coupling the GLCM with SVM raised the accuracy level to 81.7%. Yet, since precision, recall, and F1-score cannot be computed precisely, comparison is again difficult. Babu et al. [19] suggested using 2DReCA features with a Random Forest classifier, which enabled them to achieve an accuracy of 96.48%, a precision of 92.93% and a recall of 100%, with a corresponding F1-score of 96.30%. But because the system recalls everything perfectly, it could lead to more false cases. Alternatively, the hybrid features-based technique outperforms the other techniques on each of the datasets. Both the AMC and Imediatreat data achieve a maximum accuracy of 98.70%. The model also has high F1-scores, recall, and precision. With GlaS and IPC as the datasets, the method's accuracy is higher than 95% and both its performance metrics remain balanced. This finding proves that deep learning with selected handcrafted features, along with a dual-classifier structure, is extremely effective.

The proposed approach has several advantages, including its high generalization across different datasets, which suggests good adaptability and reliability in varied contexts. It also achieves balanced metrics (high precision, recall, and F1-scores), indicating its effectiveness in accurately detecting colon cancer cases while minimizing false positives and negatives. The use of comprehensive feature selection, combining texture, morphological, and statistical features, likely contributes to its high performance by capturing various aspects of the image data relevant to cancer detection. However, the approach has some limitations. Its performance may depend on the specific datasets tested, raising concerns about its robustness on unseen or real-world clinical data. The consistently high accuracy across datasets might indicate potential overfitting, especially if extensive feature engineering or tuning was conducted without robust cross-validation. Additionally, the combination of hybrid features and multiple classifiers may increase computational complexity, making it less suitable for real-time or resource-limited applications. Overall, while the proposed approach demonstrates superior performance, further validation on diverse datasets and optimization for computational efficiency would improve its practical applicability.

6 Conclusion

This study addressed the inherent challenges with conventional histopathology techniques by presenting a strong automated framework for colon cancer identification. Through a complete strategy that included smart segmentation, advanced image preprocessing, and an inventive dual-classifier system, the unique spatial arrangements of biological components in colon biopsy images were successfully assessed. The approach effectively combined Mask R-CNN and HED networks using a magnification-independent segmentation strategy, greatly improving the tissue boundary delineation. This made it possible to accurately identify intricate tissue patterns at several magnification levels, including 4X, 10X, and 40X. Normalizing tissue appearance and enabling effective feature extraction were made possible by the preprocessing methods used, such as color normalization and contrast enhancement. The feature extraction procedure was carefully planned, employing a hybrid strategy that combined morphological features and MSER approaches with texture-based and deep features derived from DTCWT. This feature set was subsequently enhanced by the addition of HOG features, which offered a thorough representation necessary for precise tissue categorization. The application of an upgraded Snake Optimization-based feature selection process that prioritized the most pertinent attributes and increased computing efficiency was a noteworthy highlight of this study. Additionally, the dual-classifier system, which combined RF and LSTM classifiers, produced notable gains in classification performance by successfully capturing both sequential data patterns and nonlinear correlations. The experiment's findings, which showed remarkable accuracy levels, confirmed the effectiveness of the presented strategy. In particular, the technique demonstrated its potential as a trustworthy tool for cancer detection by achieving an exceptional accuracy of 98.70% on the Imediatreat dataset. These developments open the door for more uniform and impartial histopathological diagnostic procedures in addition to improving the accuracy of automated cancer detection systems. This study highlights the revolutionary effects of automated techniques in medical imaging, pointing to a time when precise and effective cancer diagnosis will be the rule rather than the exception.

6.1 Limitations

Even if the system works well, it still deals with a few problems. Initially, the outcome of the experiment can be affected by unique characteristics of the dataset, for example, how the tissue was stained, captured with imaging equipment, and treated. Because of these factors, the model might not be useful outside the types of data represented in its training process. Even so, while the algorithm corrects for different amounts of zoom, it still uses ready-made datasets and may not be as accurate as hoped if the annotations are inconsistent. In the future, researchers will look at applying the model to a wide range of hospitals and using approaches that adapt it to new healthcare fields.

Future directions More research will focus on these key aspects in the future.

- Testing the proposed system on numerous hospital datasets and accessible data to check if it performs similarly in different hospitals.
- Applying methods to help your model account for errors caused by the varying colours of tissue in real histopathology, due to differences in staining techniques.
- Looking into methods that enable doctors and specialists to understand the predictions from AI-powered systems.
- Designing a system that pathologists can use either in low-resource environments or on portable devices.

Declarations

Conflict of interest The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership or other equity interest and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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